



A dual isotope approach to identify denitrification in groundwater at a river-bank infiltration site

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Abstract

The identification of denitrification in the Torgau sand and gravel aquifer, Germany, was carried out by a dual isotope method of measuring both the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- . Samples were prepared by an anion exchange resin method (Silva et al., J. Hydrol. 228 (2000) 22) with a modification to the AgNO_3 -drying process from a freeze-drying to an oven-drying method. The occurrence of denitrification in the aquifer was confirmed by comparing the reduction of dissolved oxygen, dissolved organic carbon and NO_3^- concentrations with the dual isotope signatures. High nitrate concentrations were associated with low $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values, and vice versa. The denitrification accords with a Rayleigh equation with calculated enrichment factors of $\varepsilon = -13.62\text{‰}$ for $\delta^{15}\text{N}$ and $\varepsilon = -9.80\text{‰}$ for $\delta^{18}\text{O}$. The slope of the straight-line relationship between the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ data demonstrated that the enrichment of the heavy nitrogen isotope was higher by a factor of 1.3 compared with the heavy oxygen isotope. It is concluded that the identification of this factor is a useful means for confirming denitrification in future groundwater studies.

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1. Introduction

Nitrate contamination, often related to agricultural activities, is one of the major problems in groundwater and is increasingly becoming a threat to groundwater supplies [1]. The World Health Organization has set a limit of 10 mg L^{-1} NO_3^- (as N) for drinking water to protect infants from ingesting groundwater high in NO_3^- who might otherwise develop methaemoglobinaemia (also commonly known as cyanosis and “blue baby” syndrome), a potentially fatal condition [2]. There is also substantial evidence collected from animal experiments that carcinogenic *N*-nitroso compounds are formed in the intestinal tract following ingestion of nitrate [3]. Furthermore, high concentrations of NO_3^- in river and

lake waters can cause eutrophication, often followed by fish-kills due to oxygen depletion [4].

Denitrification, the biologically mediated reduction of NO_3^- to N_2 gas, is considered to be the most important reaction that attenuates NO_3^- in groundwater [5]. Several studies have focused on understanding the factors that control denitrification in groundwater, especially studies of electron donors under several redox conditions [6–8], since this understanding contributes to the management of NO_3^- contamination in groundwater.

Several isotopic approaches have been employed to identify the presence or extent of denitrification in groundwater. The $^{15}\text{N}/^{14}\text{N}$ stable isotopic ratio in NO_3^- has been frequently measured [9], since the $\delta^{15}\text{N}$ value of the residual NO_3^- during denitrification increases as nitrate concentrations decrease. When plotting data as $\delta^{15}\text{N}$ versus the natural logarithm of the residual N-NO_3^- , the data should yield a straight line if

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denitrification is occurring, with the gradient of the line providing an estimate of the enrichment factor for the reaction, as follows:

$$\delta_R = \delta_{R0} + \varepsilon \ln C/C_0, \quad (1)$$

where δ_R is the $\delta^{15}\text{N}$ value of the reactant nitrate at time t , δ_{R0} is the initial $\delta^{15}\text{N}$ value of the nitrate, C is the NO_3^- concentration at time t , C_0 is the initial NO_3^- concentration and ε is the enrichment factor.

As a further advance of the isotopic approach, the dual isotope method, the measurement of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- , has also been employed for the evaluation of denitrification. As with $\delta^{15}\text{N}$, fractionation during denitrification causes the $\delta^{18}\text{O}$ value of the residual NO_3^- to be increased as nitrate concentrations decrease [7,5,10–12]. Similarly, the enrichment of ^{18}O in NO_3^- as a function of the natural logarithm of the residual NO_3^- yields a straight line [7]. Hence, the dual isotope method produces a distinctive isotopic signature on a $\delta^{15}\text{N}$ versus $\delta^{18}\text{O}$ cross-plot and overcomes the ambiguity associated with using only $\delta^{15}\text{N}$ data where the isotopic enrichment can be confused with human and animal waste sources of nitrogen that typically have heavy $\delta^{15}\text{N}$ signatures in excess of 10‰ [13]. However, to date, only a limited number of field studies using both $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- for the evaluation of denitrification have been reported due to the analytical difficulty of the $\delta^{18}\text{O}$ in NO_3^- measurement [14].

To overcome the analytical difficulty of measuring $\delta^{18}\text{O}$ - NO_3^- , Silva et al. [15,27] developed a convenient and reliable preparation technique, the anion exchange resin method, in order to measure both the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- in fresh water. As a result, investigations relating measurements of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- using the new technique should increase in number in the future. The main strengths of the technique are: (1) a simultaneous preparation of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- , given that the previous methodologies required separate preparations for each measurement; (2) a reduction of the laboratory preparation time of samples for $\delta^{15}\text{N}$ analysis compared to the Kjeldahl distillation technique [16]; and (3) an elimination of the highly toxic chemical usage for $\delta^{18}\text{O}$ measurements since the previous method, which was introduced by Amberger and Schmidt [17], used $\text{Hg}(\text{CN})_2$ in the preparation procedures.

In this paper, we report a minor modification to the anion resin method developed by Silva et al. [15] to further reduce the laboratory preparation time and also confirm the consistency of the modification. The modified dual isotope method was then applied at a field site in order to confirm denitrification in a previously investigated sand and gravel aquifer [18] and the results obtained compared to the few reported groundwater studies employing the dual isotope method.

2. Study area description

The study site near Torgau in eastern Germany, 247 km from the source of the River Elbe, is situated within a large basin, which is 4 km wide and 9 km long. The basin is filled with Pleistocene deposits to a depth of 10–55 m that comprise interfingering glaciofluvial sediments ranging from fine sand and silt to medium sand and gravel. The deposits are overlain by Holocene river gravels (5–8 m thick) and meadow loam (2 m thick). The Torgau aquifer can be described as a three-layer system [19]. The middle layer is formed by medium sand with a hydraulic conductivity of about $6 \times 10^{-4} \text{ m s}^{-1}$. The upper and lower layers are formed by coarse sand to gravel, with a hydraulic conductivity of about $2 \times 10^{-3} \text{ m s}^{-1}$. Two sampling profiles, Torgau A and Torgau B, constructed in 1992 were used for our studies (see Figs. 1, 2a and b).

The contribution of the upper aquifer layer to groundwater flow depends on the water level in the River Elbe, which is in direct hydraulic contact with the aquifer. Groundwater abstraction from adjacent boreholes has induced river-bank infiltration and reversed natural flow conditions within the aquifer.

The mean discharge of the River Elbe at Torgau is $340 \text{ m}^3 \text{ s}^{-1}$. Land use in the 260-m wide strip of land between the river and the production boreholes is predominantly grassland, temporarily used for sheep pasture. Land use in the remaining catchment area is dominated by arable agriculture.

The Torgau aquifer has previously been investigated regarding the highly productive river-bank infiltration scheme, especially the infiltrate flow behaviour, mixing processes, and the transport and fate of dissolved constituents that are important for improving the quality of raw waters abstracted from production boreholes [20,21].

3. Field and laboratory methods

Groundwater and river water samples were taken from cross-sections A ($n = 8$) and B ($n = 4$) (Fig. 2a and b) in May, 2001. Groundwater was either obtained using in situ membrane pumps or a mobile submersible pump lowered into 125 and 51 mm diameter observation boreholes. Membrane pumps were also used to take samples from directly below the bed of the River Elbe. Samples were collected in polyethylene bottles and stored in a cold room at 4°C until the initiation of measurements in the laboratory.

In the field, well-head measurements, including pH, alkalinity and dissolved oxygen, were conducted. The analyses of dissolved organic carbon (DOC), Cl^- , SO_4^{2-} , NO_3^- and NO_2^- were carried out in the laboratory of the Dresden University of Technology. Isotope measure-

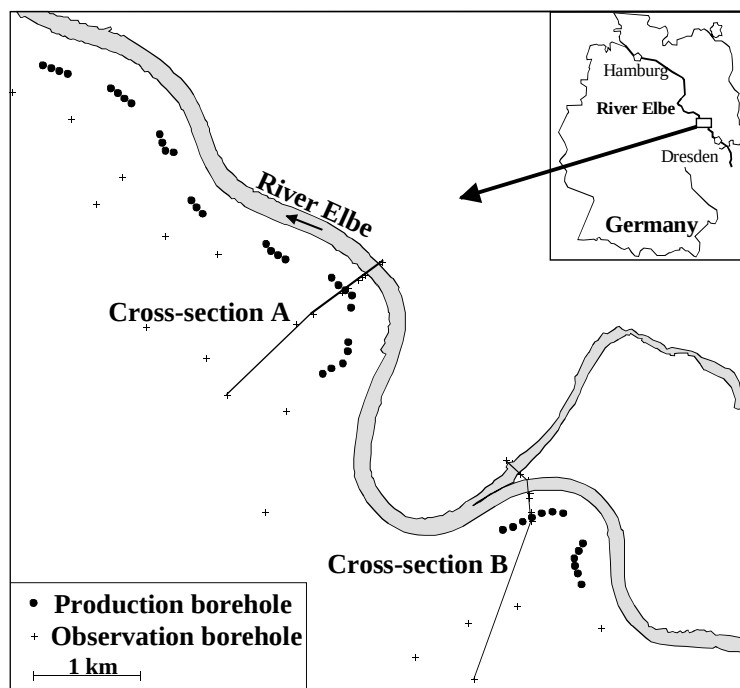


Fig. 1. Location of the study area and sampled cross-sections in the River Elbe basin near Torgau, Germany.

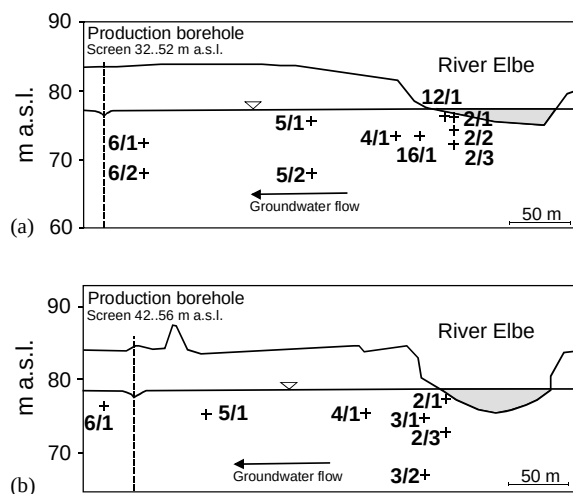


Fig. 2. Position and notation of the observation boreholes in (a) cross-section A and (b) cross-section B. Vertical depths are given in metres above sea level (m a.s.l.).

ments of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- were carried out at the University of East Anglia. Prior to analysis, water samples were filtered through a $0.45\ \mu\text{m}$ cellulose-acetate filter. Cl^- , SO_4^{2-} , NO_3^- , and NO_2^- were determined using an ion chromatograph (DX100, Dionex, Germany) with an SC-04 Hamilton PRP-X100 column (Hamilton GB Ltd, Carnforth, UK), giving an estimated detection limit of $0.05\ \text{mg L}^{-1}$ for nitrite and $0.5\ \text{mg L}^{-1}$ for the other

parameters. Samples for DOC analysis were stored at 4°C , and analysed the day after sampling with a Carbon Analyser (TOC-5000, Shimadzu, Japan) with an estimated detection limit of $0.1\ \text{mg L}^{-1}$.

A minor modification of the procedures described by Silva et al. [15,27] was developed in this study for the measurements of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- and is described as follows. First, the required amount of sample, which should contain $6.2\text{--}12.4\ \text{mg NO}_3^-$, is placed in a 1 L separatory funnel. The sample is divided if the sample volume is above 1 L. The sample is then dripped through a column filled with 2 ml of an anion exchange resin (Bio-Rad AG1-X8, 200-400 mesh anion exchange resin (Bio-Rad, Hercules, California) in chloride form) at a flow rate of $500\text{--}1000\ \text{mL h}^{-1}$. To verify that the majority of the NO_3^- is collected by the anion exchange resin, the NO_3^- concentration of the eluant is analysed by ion chromatography. Next, the sample column is placed on a stand, and 15 mL of 3 M HCl is dripped through the column in 3 mL increments to extract NO_3^- from the anion exchange resin. In a modification of the original procedure, positive air pressure, supplied by a rubber bung, is applied to the column after each 3 mL addition of HCl to remove residual eluant. The 15 mL of nitrate-bearing eluant is collected in a 100 mL glass beaker, placed in a cold water bath and then neutralised by 6 g of Ag_2O , previously rinsed in Milli-Q[®] water to remove any traces of NO_3^- . The Ag_2O is added in 1 g increments to allow the heat of

reaction to dissipate without producing a vapour. The sample is mixed between each incremental addition of Ag_2O with a flat-bottomed stirring rod to break the crust, which tends to encapsulate the unreacted reagent. A pH meter is used to verify a pH of 6–7 for the final solution. The solution is then vacuum filtered through Whatman2 #1 filters to remove AgCl precipitate and collected in a 100 mL beaker. Additional Milli-Q[®] water is used to rinse the sample nitrate through the filter. To precipitate SO_4^{2-} and PO_4^{2-} from the solution, 2 mL of 1 M BaCl_2 is added to the 100 mL beaker. After the addition of BaCl_2 , the beaker is stored at 4°C to allow the precipitate to settle. The resulting solids (BaSO_4 , BaPO_4 and AgCl) are then filtered from the solution through a 0.2 μm nylon filter.

In the next stage of preparation, the sample containing AgNO_3 is passed through a column packed with 4 mL of a cation exchange resin (Bio-Rad 100–200 mesh AG 50W X8 cation exchange resin (Bio-Rad, Hercules, California) in H^+ form) by gravity drainage. After gravity drainage, positive pressure is again applied by placing the rubber bung into the head of the column in order to blow out remaining eluant trapped in the pore space, prior to rinsing the column with Milli-Q[®] water. The sample is collected in a 50 mL glass beaker and neutralised with an excess (1 g) of Ag_2O to remove Cl^- that is added to the sample. The sample is stirred with a flat-bottomed glass rod for at least 1 min until the milky consistency diminishes leaving the solution clear. The pH is then checked to ensure it is greater than 6. The resulting AgCl precipitate and excess Ag_2O are filtered out of solution through a pre-rinsed 0.2 μm nylon filter. At this point the sample containing AgNO_3 is split into separate aliquots in the ratio 1:2 for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ analysis of NO_3^- , respectively.

At this stage, freeze-drying is the usual AgNO_3 -drying process [15]. However, in this study, an oven-drying process was employed as an alternative procedure given the melting point of AgNO_3 (212°C) and the simplicity of oven-drying compared with freeze-drying. Both aliquots for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ analyses were oven-dried at 50–60°C to evaporate water and collect AgNO_3 as a solid phase.

Nitrogen isotope analysis of the prepared AgNO_3 precipitate was conducted by conversion to N_2 gas by mixing the AgNO_3 with a CuO , Cu wire and CaO catalyst and combusting in a sealed silica tube at 850°C for 2 h using the method given by Feast and Dennis [16]. The resulting N_2 gas was cryogenically separated and analysed on a VG Isogas Sira II isotope-ratio mass spectrometer (Micromass, Manchester, UK). Oxygen isotope analysis of the prepared AgNO_3 precipitate was carried out by conversion to CO_2 gas by combustion of the AgNO_3 with 4–5 mg of finely ground spectrographic graphite at a shut-off temperature of 850°C [15]. The resulting CO_2 was cryogenically separated from other

gases also produced and analysed on the mass spectrometer.

To check the precision of the method, replicated measurements were made on solution, containing 10 mg $\text{NO}_3^- \text{L}^{-1}$, of a laboratory KNO_3 standard ($\delta^{15}\text{N} = +11.91 \pm 0.39\text{‰}$ and $\delta^{18}\text{O} = +28.44 \pm 0.29\text{‰}$). As shown in Table 1, the 1σ analytical precisions for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ analyses are $\pm 0.35\text{‰}$ and $\pm 0.69\text{‰}$, respectively, for solutions of the laboratory standard processed through the entire anion exchange resin column procedure, and confirm that the oven-drying process for recovering AgNO_3 is a satisfactory alternative to the previously reported freeze-drying method. The small differences between the laboratory standard and processed, replicated measurements of $\delta^{15}\text{N}\text{-NO}_3^-$ and $\delta^{18}\text{O}\text{-NO}_3^-$ are possibly due to small fractionation effects during the cryogenic gas separation process prior to isotope analysis. However, the systematic differences do not affect the interpretation of the nitrogen isotope chemistry in that the identification and calculation of fractionation factors caused by denitrification are unaltered. Additionally, further laboratory experiments reduced the difference to $+1.0\text{‰}$ for $\delta^{18}\text{O}\text{-NO}_3^-$ and improved the 1σ analytical precision to $\pm 0.30\text{‰}$, values that are comparable with previous studies [22,15].

A further methodological development for samples containing DOC was to use silicalite in preference to graphite as employed in previous studies [15]. Graphite has the disadvantage of removing both DOC and some NO_3^- by absorption and so possibly causing

Table 1

Replicate isotope analyses of the laboratory standard KNO_3 solution prepared using the modified anion exchange resin method

Solution no.	$\delta^{15}\text{N}\text{-NO}_3^-$ (‰)	$\delta^{18}\text{O}\text{-NO}_3^-$ (‰)
1	11.64	24.81
2	12.35	25.48
3	12.41	25.23
4	12.14	26.02
5	12.33	26.34
6	12.52	26.95
7	12.41	26.42
8	12.60	26.55
9	12.91	25.62
10	11.93	25.23
	$\bar{x} = 12.32$	$\bar{x} = 25.87$
	$\sigma = 0.35$	$\sigma = 0.69$

Note: Isotope ratios are defined using the δ -notation as: $\delta X(\text{‰}) = (R_S/R_{\text{ST}} - 1) \times 1000$, where X is ^{15}N or ^{18}O and R is the $^{15}\text{N}/^{14}\text{N}$ or $^{18}\text{O}/^{16}\text{O}$ ratio of the sample (S) and reference standard (ST), respectively. The reference standards are Air for $\delta^{15}\text{N}$ and Vienna Standard Mean Ocean Water (VSMOW) for $\delta^{18}\text{O}$.

fractionation affects with respect to nitrogen and oxygen isotopes. Laboratory experiments showed that complete removal of DOC was possible without a reduction in the NO_3^- concentration of the sample. Further discussion of the methodology and the laboratory results are given in Hwang et al. [23].

4. Results and discussion

4.1. Water chemistry

Table 2 presents the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ isotopic data, the concentrations of anions (Cl^- , SO_4^{2-} , NO_3^- and NO_2^-), DOC, dissolved oxygen (O_2) and alkalinity and pH. The data in Table 2 mainly represent samples from levels 1 and 2 in the upper aquifer layer covering a distance of 300 m from the river. This region of the aquifer represents infiltrated river water that does not mix with groundwater from other sources [19]. However, sampling points A5/1 and A6/1 may contain a small amount of seepage water from the soil surface, but the contribution to the ionic concentrations is variable depending on the hydraulic conditions prior to the sampling dates. The changes in groundwater ion concentrations and changing groundwater flow paths in this aquifer entirely depend on production borehole pumping rates and river water levels, and it is therefore less straightforward to identify a straight flow path attributable to river water input and to predict changes in anion concentrations.

The decrease of dissolved oxygen from 11.5 mg L^{-1} in the River Elbe to $<1 \text{ mg L}^{-1}$ in observation boreholes A2/3 and B2/3 in the infiltrating river water is observed

as the river water infiltrates the river-bed sediments. Simultaneously, reductions in DOC of $0.7\text{--}1.0 \text{ mg L}^{-1}$ are observed at the closest points to the river, A2/1 and B2/1, with DOC concentrations at all other points found to be lower than in the River Elbe. Also, a reduction in the NO_3^- concentration of the infiltrated river water is observed to occur at most of the sampling points in the aquifer once the river water infiltrates the river-bed sediments. These trends, particularly the reduction in dissolved oxygen, DOC and NO_3^- near the river bed illustrate the beneficial effect of the colmation layer for bank filtration schemes in attenuating introduced contaminants [24]. The decrease in the concentrations of DOC and NO_3^- in the infiltrating groundwater is caused by the bacterially mediated reduction of nitrate and confirms the previous findings of Grischek et al. [18] who presented mass-balance calculations of the denitrification process involving both dissolved and solid organic carbon fractions as the e^- donors in the river bed and aquifer material. NO_2^- is an unstable intermediate product in the denitrification reaction and concentrations of less than 0.6 mg L^{-1} are recorded in Table 2.

In the present study, there are three points, A5/1, A6/1 and B3/2, where the NO_3^- concentrations are above that in the River Elbe. The concentration of SO_4^{2-} is also distinctly higher than for the River Elbe at sampling points A5/1 and A6/1. The high major anion concentrations at these two points are most likely due to the alteration of hydraulic conditions prior to the sampling date. The unsaturated zone above points A5/1 and A6/1 was partly flushed during a period of high water level in the river 1 month before sampling. As the water level descended after this period, this resulted in a

Table 2
Hydrochemical and stable isotope data for groundwater samples from the Torgau aquifer, Germany, collected in May 2001

No.	$\delta^{15}\text{N}$ (‰)	$\delta^{18}\text{O}$ (‰)	Cl^- (mg L^{-1})	SO_4^{2-} (mg L^{-1})	NO_3^- (mg L^{-1})	NO_2^- (mg L^{-1})	DOC (mg L^{-1})	DO (mg L^{-1})	pH	Alk. (mg L^{-1})
A2/1	14.10	13.82	20.5	56.3	13.9	0.56	5.3	1.4 ^a	7.1	104
A2/2	14.38	14.72	21.5	58.9	14.7	0.58	5.5	1.0 ^a	7.0	103
A2/3	15.58	15.62	22.0	59.9	15.2	0.42	5.1	0.5 ^a	7.0	101
A4/1	11.62	12.00	23.1	61.4	14.9	<0.05	4.2	3.5	6.8	104
A5/1	11.17	9.22	25.5	75.2	21.8	<0.05	3.5	4.6	6.8	86
A6/1	16.64	12.69	32.4	74.4	19.2	<0.05	2.9	3.1	6.8	101
A12/1	14.19	14.22	25.6	69.9	14.0	0.64	4.8	1.0 ^a	7.2	114
A16/1	20.60	16.78	24.6	64.6	8.8	0.32	5.5	<0.2	7.0	128
B2/1	17.01	16.60	25.6	69.4	10.6	0.52	5.6	1.6 ^a	7.1	115
B2/3	14.39	11.53	27.7	69.8	15.5	0.37	5.6	0.5 ^a	7.0	109
B3/1	14.98	12.98	25.6	68.5	13.0	0.55	5.4	1.7 ^a	7.1	116
B3/2	15.92	11.68	27.6	68.1	17.1	0.16	4.4	3.0 ^a	7.3	111
River Elbe	8.85	8.02	26.8	68.2	16.8	0.14	6.3	11.5	7.9	92

Note: See Fig. 2 for location of sample sites (Alk. = Alkalinity).

^a overestimation possible due to low sampling volume of in-situ membrane pumps.

contribution to the high major anion concentrations observed at these two points.

Grischek et al. [18] reported that sampling points A5/1 and A6/1 are influenced by an additional N-source through recharge from the soil surface in an area of former sheep pasture and manure application near the observation boreholes. This source can also be identified by the stable isotope data. According to previous studies [25,26], the $\delta^{15}\text{N}$ - NO_3^- composition in aquatic systems derived from animal waste sources is generally in the range of +10‰ to +20‰. The $\delta^{15}\text{N}$ of NO_3^- signatures at A5/1 and A6/1 are +11.17 and +16.64‰, respectively, and are therefore consistent with the range expected for animal waste-impacted groundwater.

The $\delta^{18}\text{O}$ of NO_3^- values also support the above explanation, especially A5/1. Kendall [4] mentioned that the $\delta^{18}\text{O}$ of NO_3^- newly formed from in situ nitrification of ammonium should be in the range of -10‰ to +10‰. Hence, the $\delta^{18}\text{O}$ of NO_3^- value of +9.22‰ at A5/1 indicates that the dominant process at A5/1 is the nitrification of ammonium, with a probable source in the animal waste. However, the $\delta^{18}\text{O}$ of NO_3^- value of +12.69‰ at A6/1 is higher than expected for in situ nitrification of ammonium, but several studies [5,17] have reported similar or even more enriched values in a nitrification-dominant situation. Several explanations for this were presented previously [4], but, in our study, episodic denitrification in isolated pores that temporarily become anoxic could also be responsible for the enriched values, considering the potential for denitrification in the aquifer at this bank filtration site.

The situation at sampling point B3/2 is influenced by the relatively older groundwater at this position (mean residence time of about 50 days) compared with groundwater in the region above. As a result, groundwater sampled at point B3/2 represents water that infiltrated the sediments and flowed into the aquifer during a period of higher NO_3^- , SO_4^{2-} and Cl^- concentrations (Cl^- concentrations in the River Elbe were above 28 mg L⁻¹ before mid-March). The lower temperature of the river water in the period before mid-March may also have limited denitrification in the river bed resulting in the relatively high NO_3^- concentration of 17.1 mg L⁻¹ recorded at point B3/2 during this study.

Given the uncertain conditions explained above, the three samples A5/1, A6/1 and B3/2 are excluded from further discussion of the dual isotope approach to demonstrate denitrification along the cross-sections sampled in this study.

4.2. $\delta^{15}\text{N}$ in nitrate

The range of measured $\delta^{15}\text{N}$ - NO_3^- observed in the specific area below the river bed of the River Elbe is between +8.85‰ (River Elbe) and +20.60‰ (point 16/1) [median value = +14.39‰ ($n = 13$), Table 2]. The extent

of $\delta^{15}\text{N}$ enrichment of the residual NO_3^- due to denitrification is not as large as found in other denitrification studies, such as +8.40‰ to +31.20‰ in the Chalk aquifer of northern France [9], +6.00‰ to +58.30‰ in a sand aquifer in Southern Ontario, Canada [5] and +5.00‰ to +36.00‰ in a sand aquifer in Ontario, Canada [12].

Although the sampling points in this study are not identical to those sampled in the previous study [18], the trend of $\delta^{15}\text{N}$ values in NO_3^- in both studies is very similar. The $\delta^{15}\text{N}$ - NO_3^- values of river-bank infiltrate demonstrate an enrichment with distance from the river at cross-section A, while the NO_3^- concentrations show an apparent decrease (Table 2, Fig. 3). Although the $\delta^{15}\text{N}$ values in NO_3^- are enriched with a reduction of NO_3^- concentration at cross-section B, the enrichment does not correlate with distance from the river. This is likely to be due to different starting concentrations of nitrate and temperature at the time of infiltration, the varying age of groundwater within the aquifer, which ranges from a couple of days (A2/1, B2/1) to more than 60 days (A6/1) and dispersion of the river-bank infiltrate under varying hydraulic conditions.

Overall, the $\delta^{15}\text{N}$ versus $\ln(\text{N}-\text{NO}_3^-)$ plot for the Torgau aquifer (Fig. 3) shows a clear trend of $\delta^{15}\text{N}$ enrichment correlated with decreasing NO_3^- concentration, providing strong evidence for denitrification as the mechanism causing nitrate attenuation at this site. Calculation of the isotopic enrichment factor (ϵ) using Eq. (1) gives a value of -13.62‰ and is almost the same as that calculated in the previous study, -14.6‰ [18]. These values are within the range of isotopic enrichment factors reported for denitrification in various aquatic studies, $\epsilon = -40$ ‰ to -5‰ [4].

4.3. $\delta^{18}\text{O}$ in nitrate

The range of measured $\delta^{18}\text{O}$ - NO_3^- values determined for this study is between +8.02‰ (River Elbe) and +16.78‰ (point 16/1) [median value = +12.98‰ ($n = 13$), Table 2]. Fig. 4 demonstrates that the $\delta^{18}\text{O}$ in the residual NO_3^- is enriched as the NO_3^- concentration is reduced. There is a clear $\delta^{18}\text{O}$ enrichment trend between the River Elbe and sampling point A16/1 at cross-section A correlating with distance from the river. However, and as noted for the $\delta^{15}\text{N}$ data, there is no obvious correlation with distance from the river and the enrichment in $\delta^{18}\text{O}$ for the observed sampling points at cross-section B. Again, these differences in the $\delta^{18}\text{O}$ trends at cross-sections A and B are likely to be due to hydraulic effects, different conditions at the time of infiltration and differences in the age of groundwater within the aquifer, especially at cross-section B.

It is reported that denitrification causes an increase in the $\delta^{18}\text{O}$ of the residual NO_3^- due to isotopic fractionation [3], but there are few comparable studies of $\delta^{18}\text{O}$

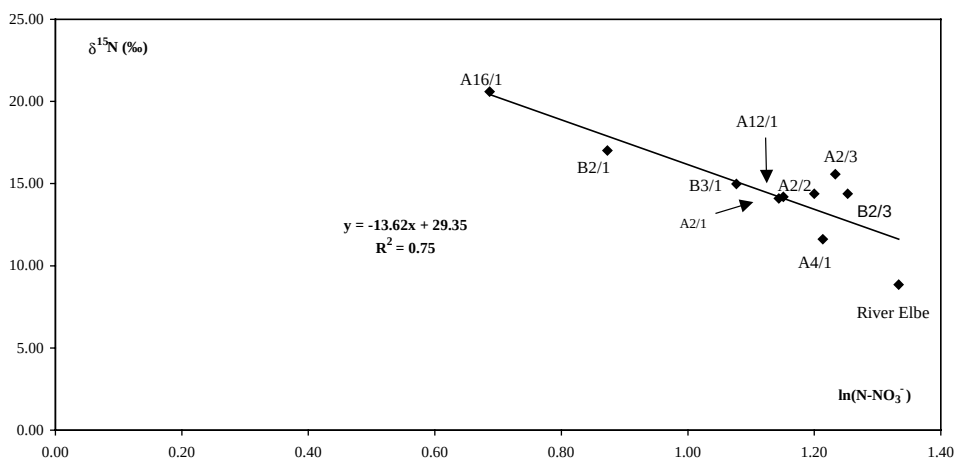


Fig. 3. Cross-plot of the $\delta^{15}\text{N}$ composition of dissolved NO_3^- versus the natural log of N-NO_3^- concentrations (in mg L^{-1}) of samples from the Torgau aquifer, Germany (without samples A5/1, A6/1 and B3/2).

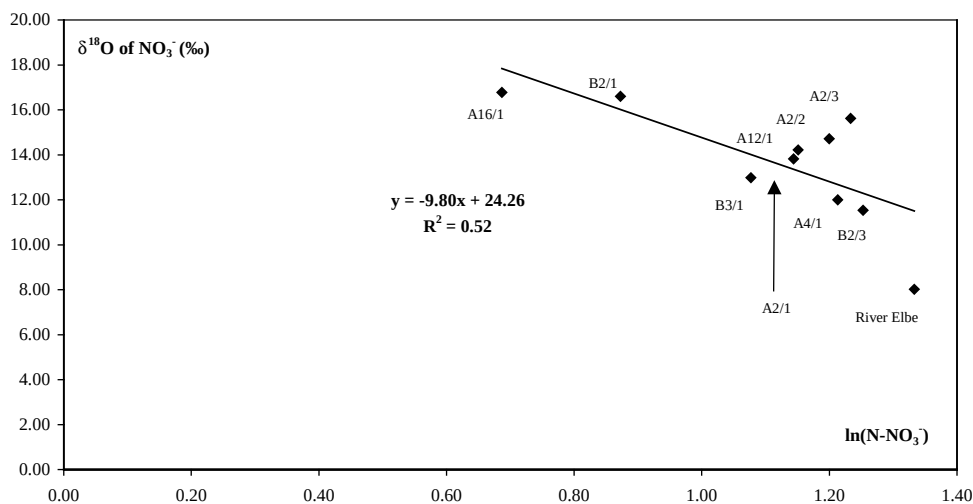


Fig. 4. Cross-plot of the $\delta^{18}\text{O}$ composition of dissolved NO_3^- versus the natural log of N-NO_3^- concentrations (in mg L^{-1}) of samples from the Torgau aquifer, Germany (without samples A5/1, A6/1 and B3/2).

isotope enrichment. The reported extent of $\delta^{18}\text{O}$ enrichment of residual NO_3^- due to denitrification was from +3.08‰ to +55.93‰ in a sand aquifer in Hannover, Germany [7], −0.5‰ to +16.6‰ in sand aquifers in Southern Ontario, Canada [5,10] and +1.2‰ to +12.7‰ in a sand aquifer in Ontario, Canada [12]. These differences in the range of enrichment are most likely influenced by the specific conditions of each system, e.g. substrate concentration, concentration and microbial availability of electron donors, temperature, denitrification rate, type of bacterium involved and growth conditions.

Although there are only a few studies that calculate an isotopic enrichment factor for $\delta^{18}\text{O}$ in the residual NO_3^- during denitrification, it is often difficult to calculate an

enrichment factor from field evidence due to a lack of parameter values, for example the initial substrate concentration. The isotopic enrichment factor can be estimated using Eq. (1) by finding the slope of the $\delta^{18}\text{O}$ versus $\ln(\text{N-NO}_3^-)$ plot. For this study, the calculated isotopic enrichment factor, ϵ , found from the slope of Fig. 4 is −9.80‰, and is close to the value of −8.0‰ reported by Böttcher et al. [7] and moderately higher than the value of −18.3‰ reported by Mengis et al. [11].

4.4. $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in nitrate

The observed linear relationships between the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values and the reduction in the natural logarithm of NO_3^- concentration (Figs. 3 and 4) confirm

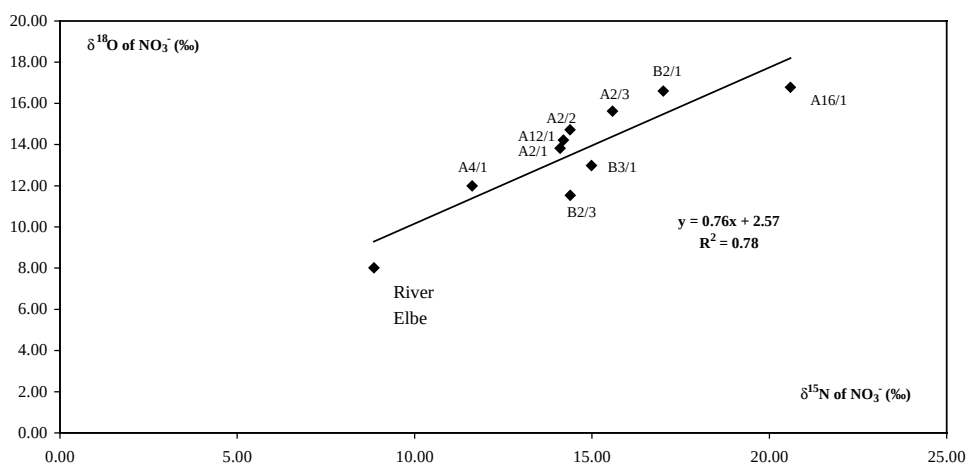


Fig. 5. Cross-plot of the $\delta^{18}\text{O}$ composition of dissolved NO_3^- versus the $\delta^{15}\text{N}$ composition of dissolved NO_3^- of samples from the Torgau aquifer, Germany (without A5/1, A6/1 and B3/2).

the occurrence of microbial denitrification at the Torgau aquifer site. This being the case, then the denitrification reaction should produce a distinctive isotope signature on a cross-plot of $\delta^{15}\text{N}$ versus $\delta^{18}\text{O}$, as shown in Fig. 5. The points in the lower left corner of Fig. 5 correspond to water samples with high NO_3^- concentrations, whereas the points with increasing $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values correspond to samples with reduced NO_3^- concentrations. The slope of the line in Fig. 5 demonstrates that the enrichment for the heavy nitrogen isotope, ^{15}N , is higher by a factor of 1.3 than for the heavy oxygen isotope, ^{18}O . This value of 1.3 is similar to the four reported results available from field studies, giving fractionation factors of 1.5 [11], 1.7 [10], 1.8 [12] and 2.1 [7,5]. Thus, although the individual isotopic enrichment factors for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- during denitrification may vary according to the specific field conditions, the fractionation factor of $\delta^{15}\text{N}/\delta^{18}\text{O}$ in NO_3^- appears to remain constant. This preferential enrichment of ^{15}N compared to ^{18}O might be explained by either the difference in the relative strength of the $^{15}\text{N}\text{--}^{16}\text{O}$ and $^{14}\text{N}\text{--}^{18}\text{O}$ bonds or the involvement of ^{17}O in the process.

5. Conclusions

The application of the dual isotope method of measuring both the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ compositions of dissolved NO_3^- in this study confirmed the occurrence and importance for water supply of denitrification in the Torgau aquifer. The denitrification reaction resulted in the enrichment of ^{15}N and ^{18}O of the residual NO_3^- in the infiltrating river water with isotopic enrichment factors (ϵ) of -13.62‰ for N and -9.80‰ for O. The denitrification reaction also produced a distinctive linear isotopic relationship on a $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ cross-plot

demonstrating a preferential enrichment of the heavy nitrogen isotope by a factor of 1.3 compared with the heavy oxygen isotope. This value of 1.3 is similar to the few previously reported values, and it is concluded that this is a useful means for confirming denitrification in future groundwater studies.

Currently, we recommend the anion exchange resin method as a suitable method for preparing samples for measurements of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- , since the method allows the simultaneous preparation of the sample for both isotope measurements and does not require any poisonous chemicals. The applicability of an oven-drying process for the preparation of the AgNO_3 precipitate was proved in this study, such that the more usual, but longer freeze-drying process is considered to be unnecessary.

Although the anion exchange resin method is currently a suitable method for measuring $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in NO_3^- , further developments are required. For example, the cost of Ag_2O is expensive for processing large numbers of samples [15]. A simple methodology for regenerating Ag_2O from AgCl waste is required to tackle this problem.

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